

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 3

Duration :60 Mins
Off Class
Total Marks:30

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I H. VAISHNAV (192521251) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CO2 - AT-3 (Compiler Design)

Q1.

Streaming XML Data Parsing :-

Grammar :-

$$S \rightarrow \langle id \rangle S \mid \langle /id \rangle S \mid \epsilon$$

(i) Explain how nested structures are validated

The grammar validates nesting recursively. Every opening tag $\langle id \rangle$ must have a matching closing tag $\langle /id \rangle$.

(ii) Demonstrate parsing :-

Input : $\langle id \rangle \langle id \rangle \langle /id \rangle \langle /id \rangle$

Leftmost Derivation :-

$$\begin{aligned} S &\Rightarrow \langle id \rangle S \langle /id \rangle S \\ &\Rightarrow \langle id \rangle \langle id \rangle S \langle /id \rangle S \langle /id \rangle S \\ &\Rightarrow \langle id \rangle \langle id \rangle \epsilon \langle /id \rangle \epsilon \langle /id \rangle \epsilon \\ &\Rightarrow \langle id \rangle \langle id \rangle \langle /id \rangle \langle /id \rangle \end{aligned}$$

Thus, the input is accepted.

(iii) Handle incomplete input

Example :-

$\langle id \rangle \langle id \rangle \langle /id \rangle$

The parser detects that the outer $\langle id \rangle$ tag is not closed.

(iv) Identify mismatched tags

Example :- $\langle id \rangle \langle id \rangle \langle id \rangle \langle name \rangle$

(v) Suggest streaming parser design

use stack - based parsing,
push opening tags onto the stack.

22. Notation control :- system - logical
Expression Parsing.

(i) The grammar gives higher
preference to AND than OR, ensuring
accurate interpretation of flight rules.

(ii) Demonstration :-

E

$\Rightarrow T E'$

$\Rightarrow \text{OR } T' E'$

$\Rightarrow \text{condition } T' E'$

$\Rightarrow \text{condition and OR } T' E'$

$\Rightarrow T' E'$

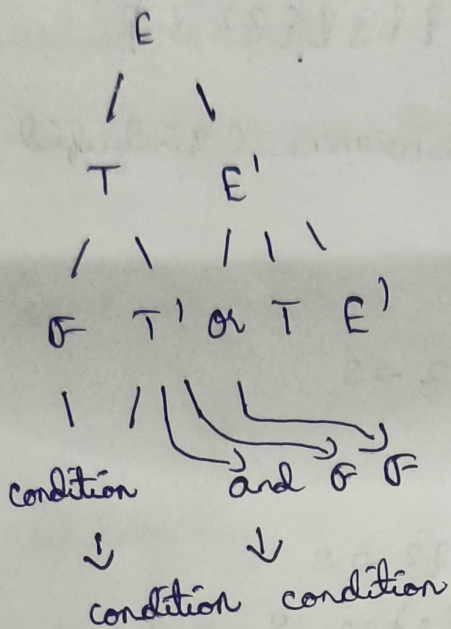
$\Rightarrow E'$

$\Rightarrow T E'$

$\Rightarrow \text{OR } T' E'$

$\Rightarrow \text{condition and condition or
condition.}$

(iii) Construct parse tree :-



(iv) Show risk of ambiguity :-

This can produce different decisions, causing safety risks in aviation control.

Q3. Blockchain Network - Nested Block validation.

dyganner : $S \rightarrow \{S\} SLE$

Input : 2 2 3 2 2 3 3 3

(i) Unested blocks are allowed because S appears inside the production itself.